

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

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b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

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2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

15m: A^n 의 rank, $n=2$ 일 때부터 시작

a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

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3. Find all $a, b, c \in \mathbb{R}$ such that

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$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

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4, Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol), Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B)^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \mathcal{Q}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 라고 시작하고, hom 임을 보려면 $\mathcal{Q}$ 가 $\mathbb{R}^{\times}$ 에 닫혀 있는지 보자!

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5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b). By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So,

$$12x \equiv 15y \pmod{21}$$

has distinct $\gcd(12, 21) = 3$ solutions. $\square$

